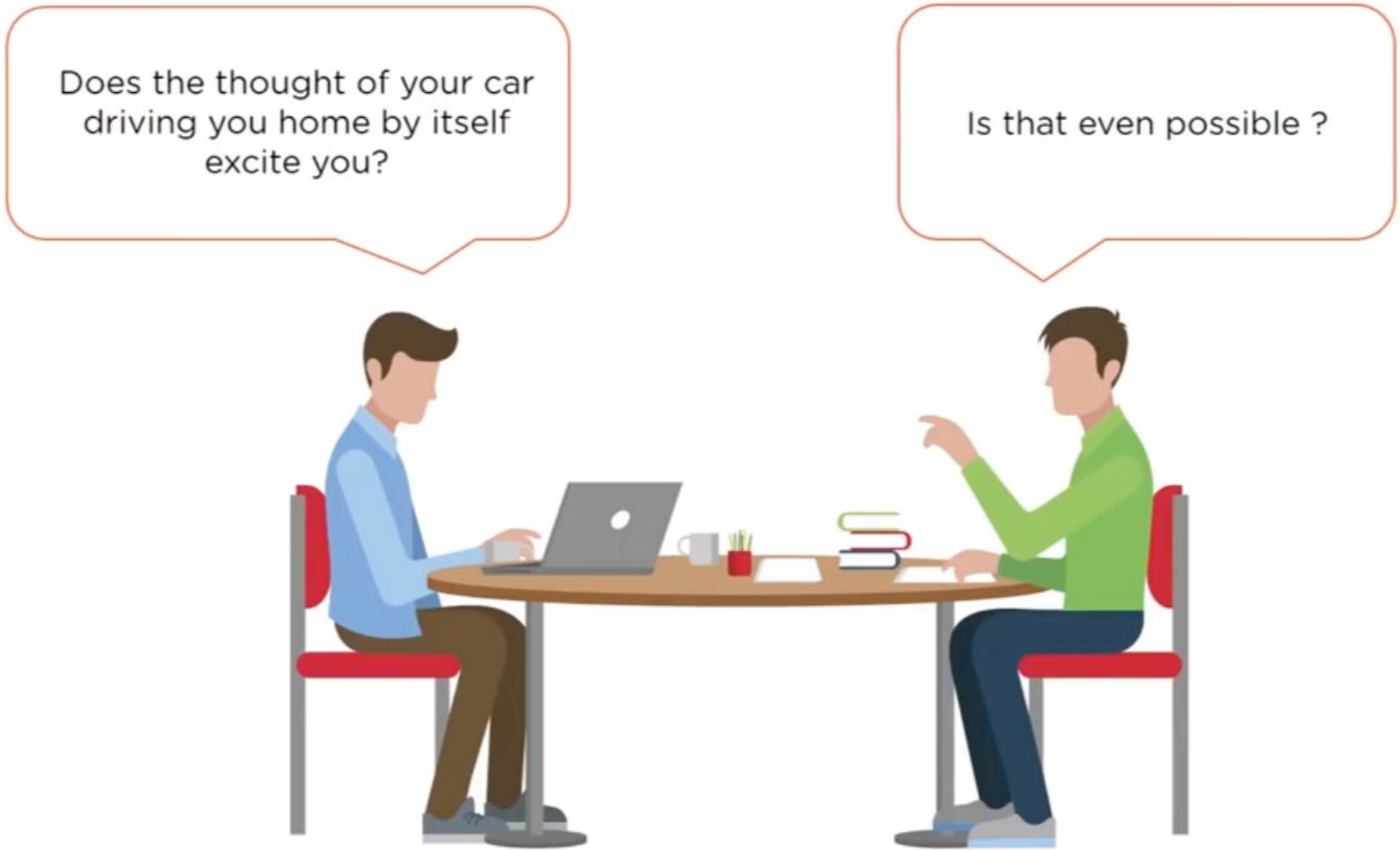
# CSE315:Introduction to Data Science

WEEK-1

# Need For Data Science




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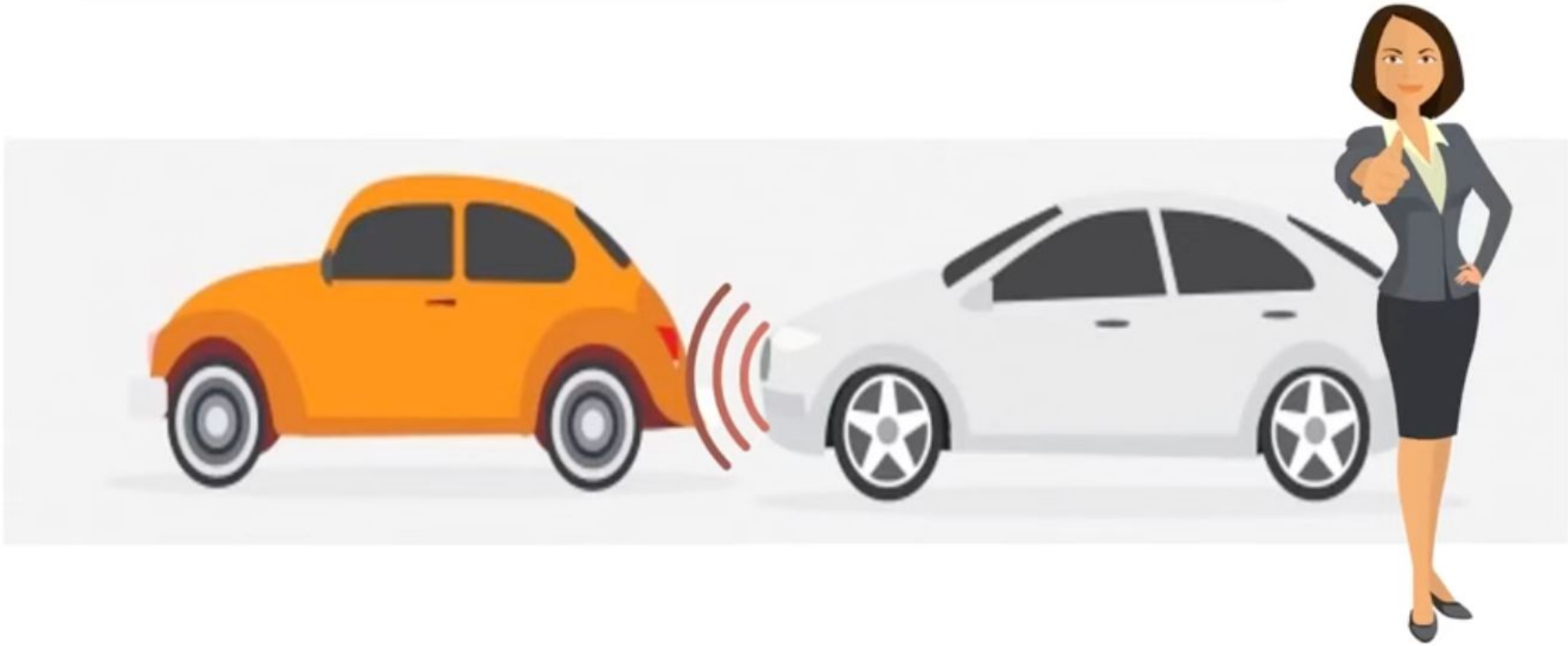
# Need For Data Science

TESLA

Baidu 百度

WAYMO

Self driving cars will root out more than 2 million deaths caused by car accidents annually.



# Need For Data Science

Dear Flyer, We regret to inform you that your flight has been cancelled due to delay from Airbus on account of engine delivery



1. Due to lack of data available, flights are often delayed or cancelled at the last minute
2. Due to improper route planning, customers don't get the flight for desired time and duration
3. Incorrect decisions in selection of right equipment leads to unplanned delays and cancellations

# Need for data Science

Using Data Science, we can achieve the following:

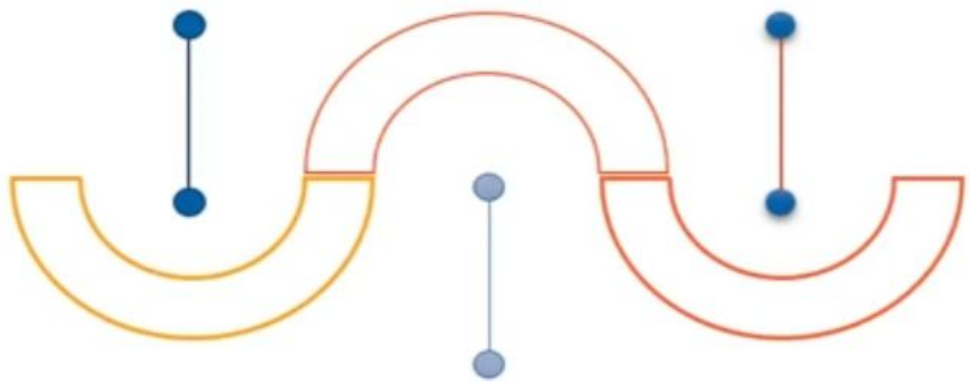
- Route Planning: Whether to schedule direct or connecting flights
- Predictive analytics model can be built to foresee flight delays
- Promotional offers depending on customer booking patterns
- Deciding which class of planes to purchase for better performance



Logistics companies like FedEx are using Data Science models for operational efficiency

Discover the best routes to ship

The best suited time to deliver





The best mode of transport

So Data Science is mainly needed for:



**Better Decision Making**

Whether A or B?



**Predictive Analysis**

What will happen next?

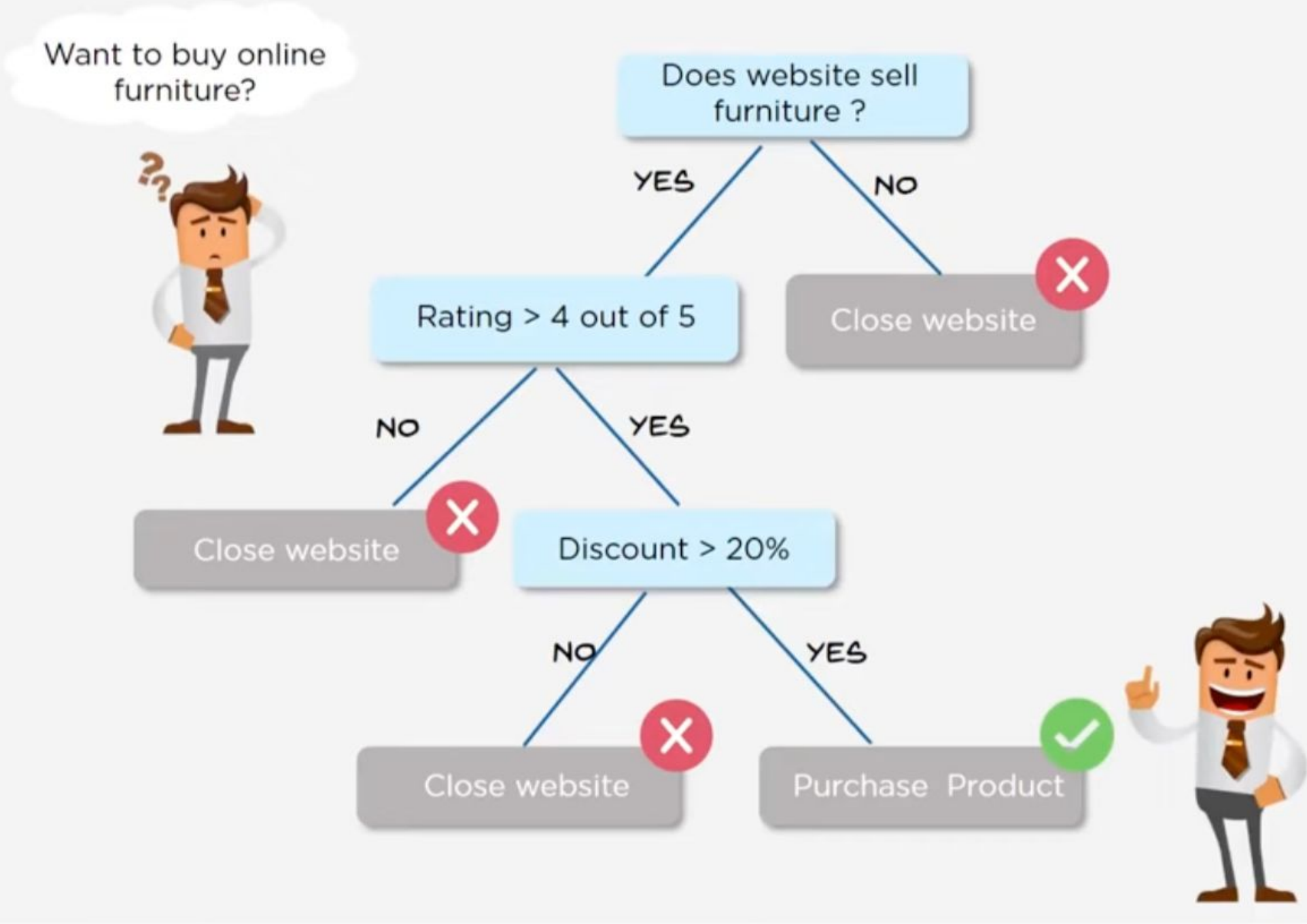


**Pattern Discovery**

Is there any hidden information in the data?

# Need for data Science




Want to buy online furniture?
Does website sell furniture ?
YES
NO
Rating > 4 out of 5
Close website
NO
YES
Close website
Discount > 20%
NO
YES
Close website
Purchase Product

# Steps involved in Data Science



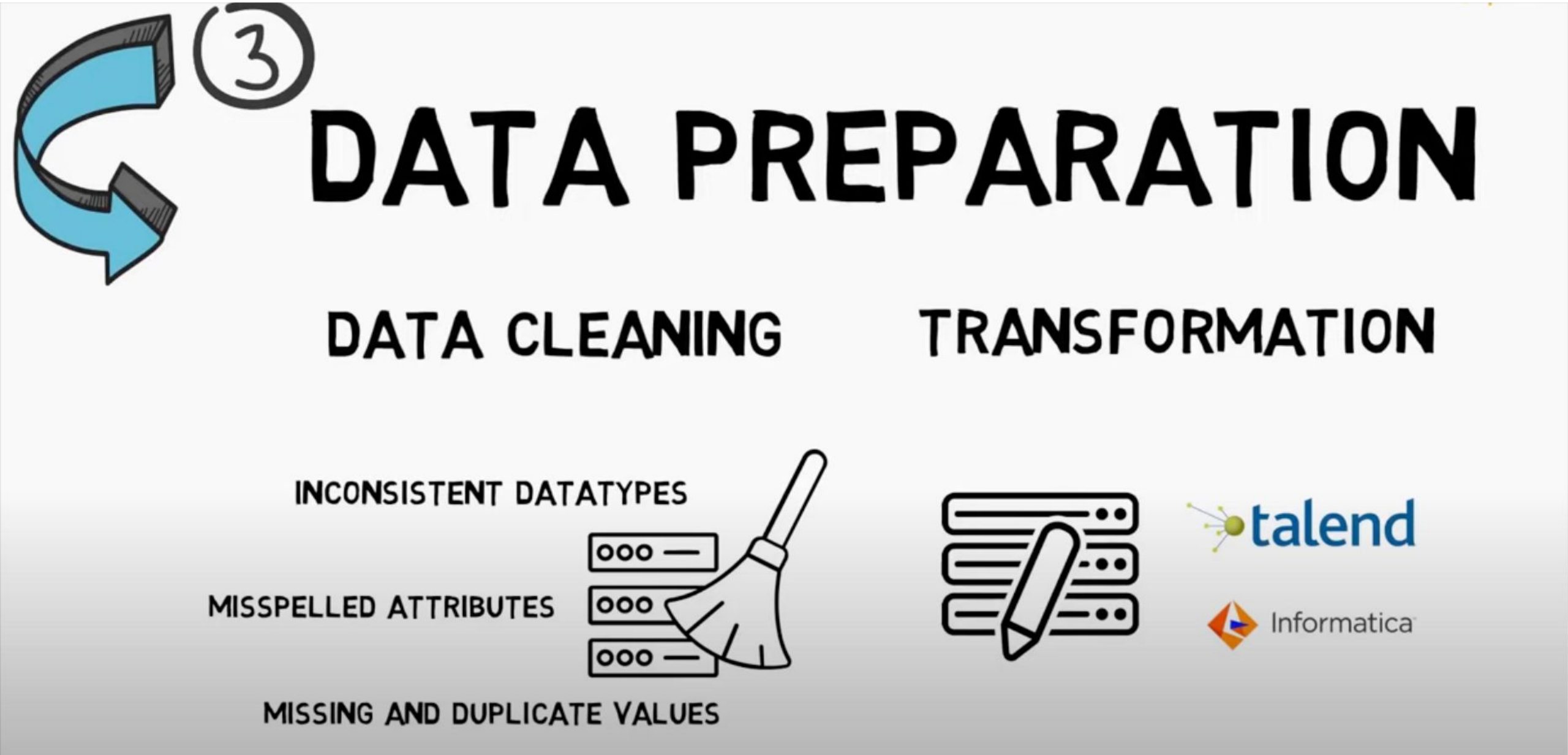
# Steps involved in Data Science



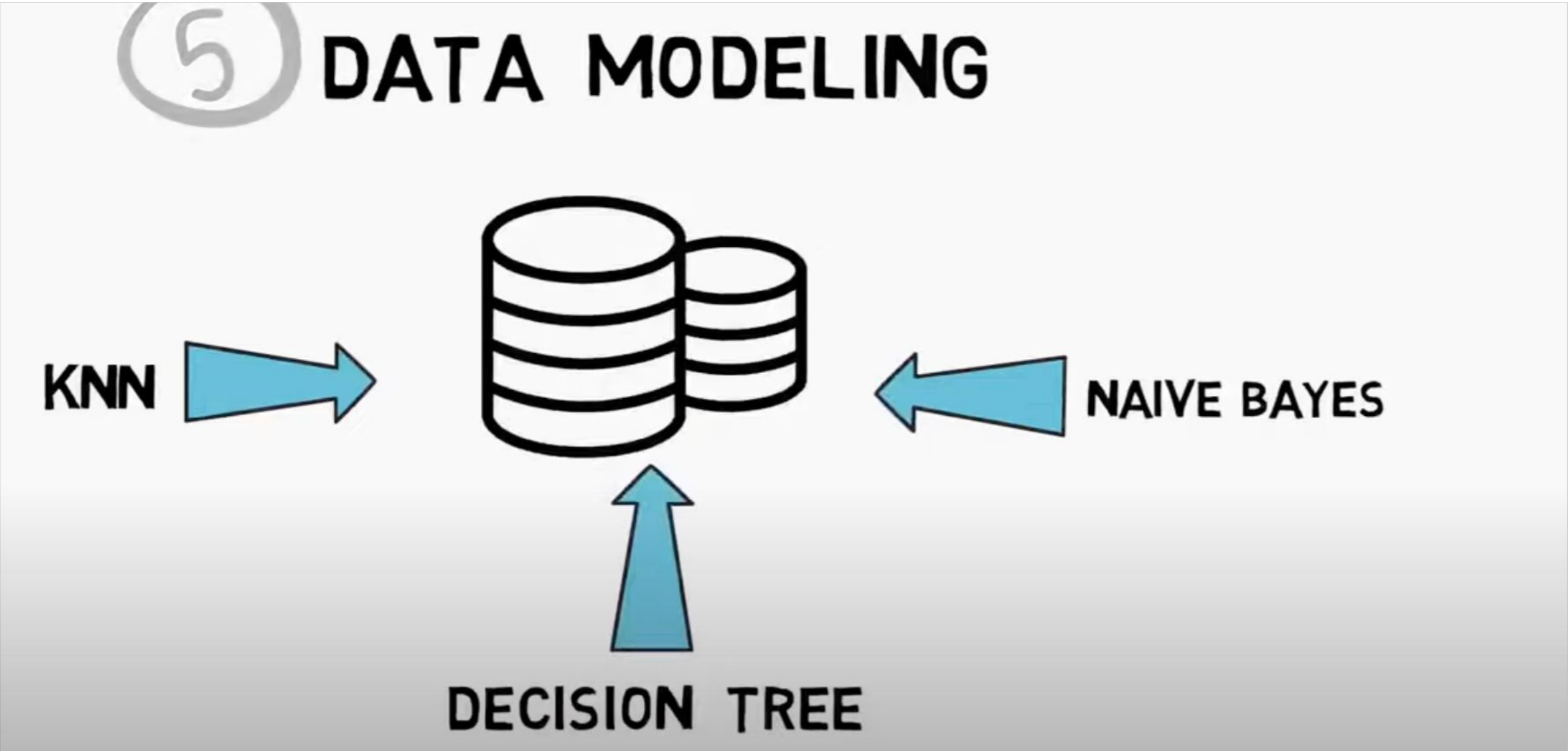

## ② DATA ACQUISITION

- WEB SERVERS
- LOGS
- DATABASES
- API'S
- ONLINE REPOSITORIES

# Steps involved in Data Science



# Steps involved in Data Science



# Steps involved in Data Science




IDENTIFY THE MODEL THAT BEST FITS THE BUSINESS REQUIREMENT

TRAINS THE MODELS ON THE TRAINING DATASET AND TEST

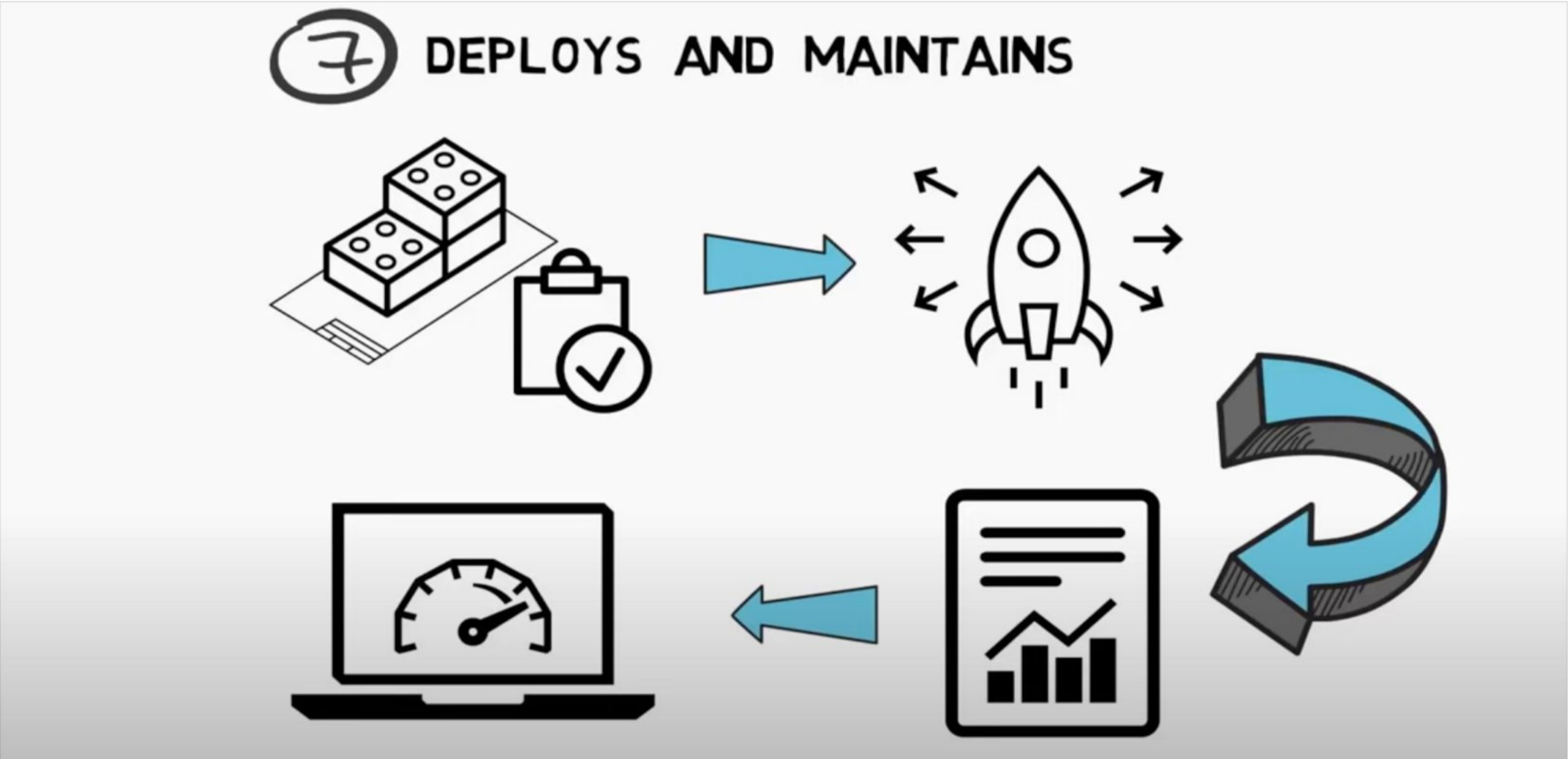
SELECT THE BEST PERFORMING MODEL

python™

R

sas

# Steps involved in Data Science



# Data Science

- Data science is a collection of techniques used to extract value from data. It has become an essential tool for any organization that collects, stores, and processes data as part of its operations.
- Data science techniques rely on finding useful patterns, connections, and relationships within data. Being a buzzword, there is a wide variety of definitions and criteria for what constitutes data science.
- Data science is also commonly referred to as knowledge discovery, machine learning, predictive analytics, and data mining.

# Data Science

- Data science starts with data, which can range from a simple array of a few numeric observations to a complex matrix of millions of observations with thousands of variables.
- Data science utilizes certain specialized computational methods in order to discover meaningful and useful structures within a dataset.
- The discipline of data science coexists and is closely associated with a number of related areas such as database systems, data engineering, visualization, data analysis, experimentation, and business intelligence (BI).

# AI, ML and Data Science

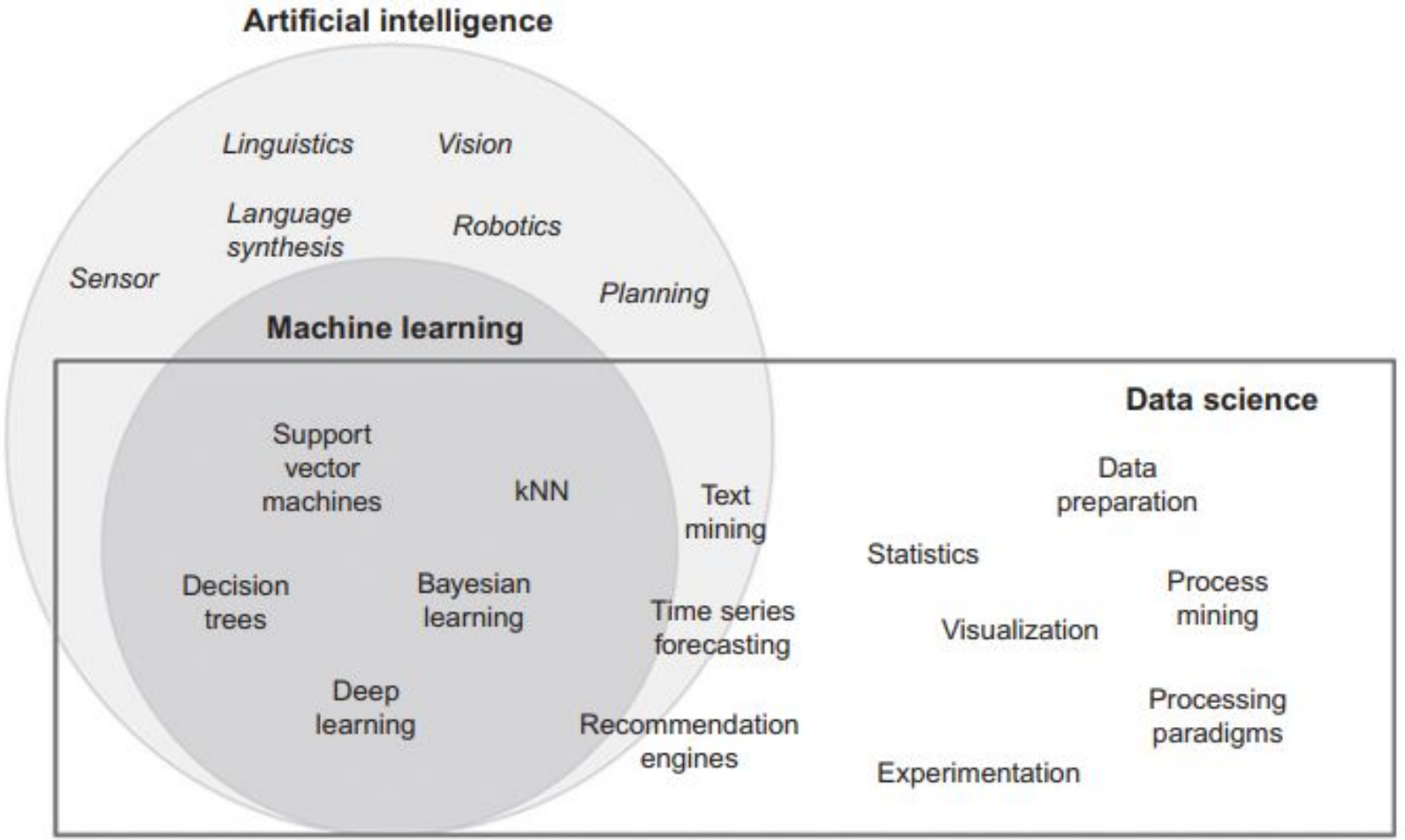
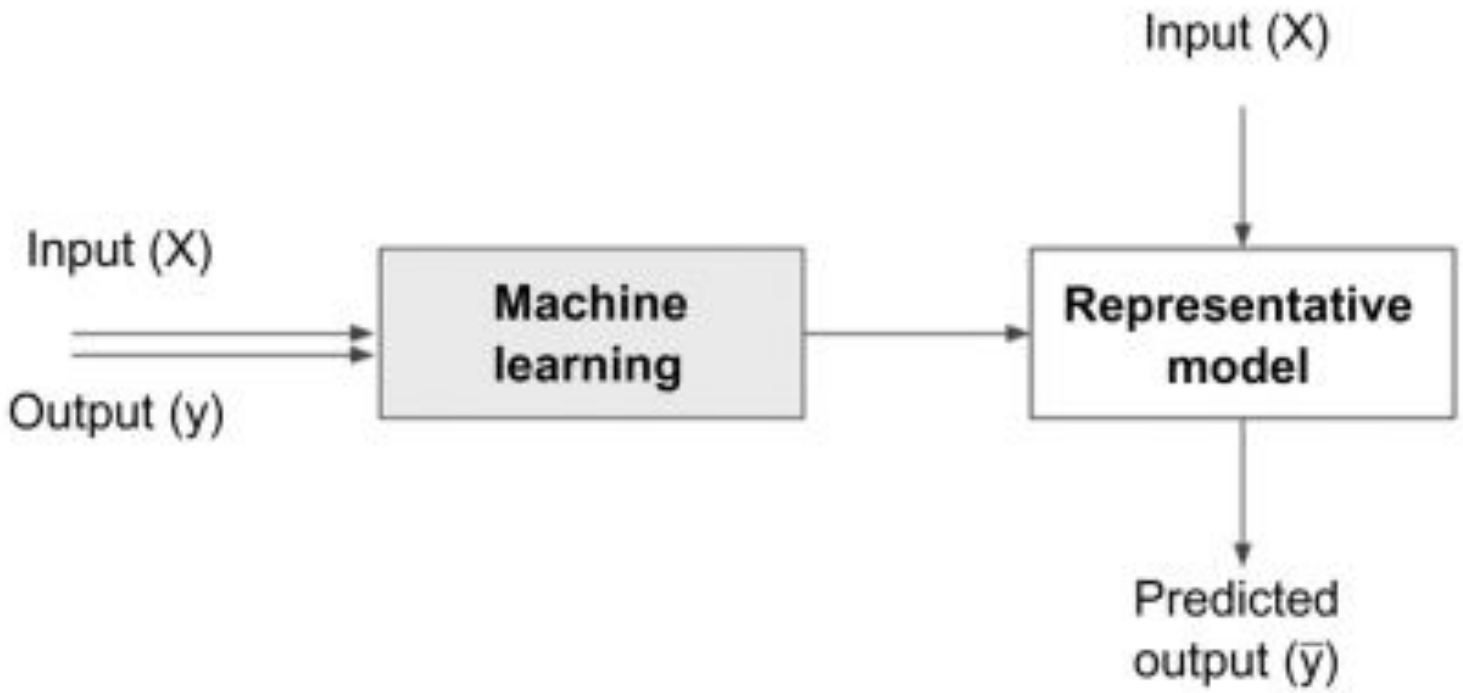
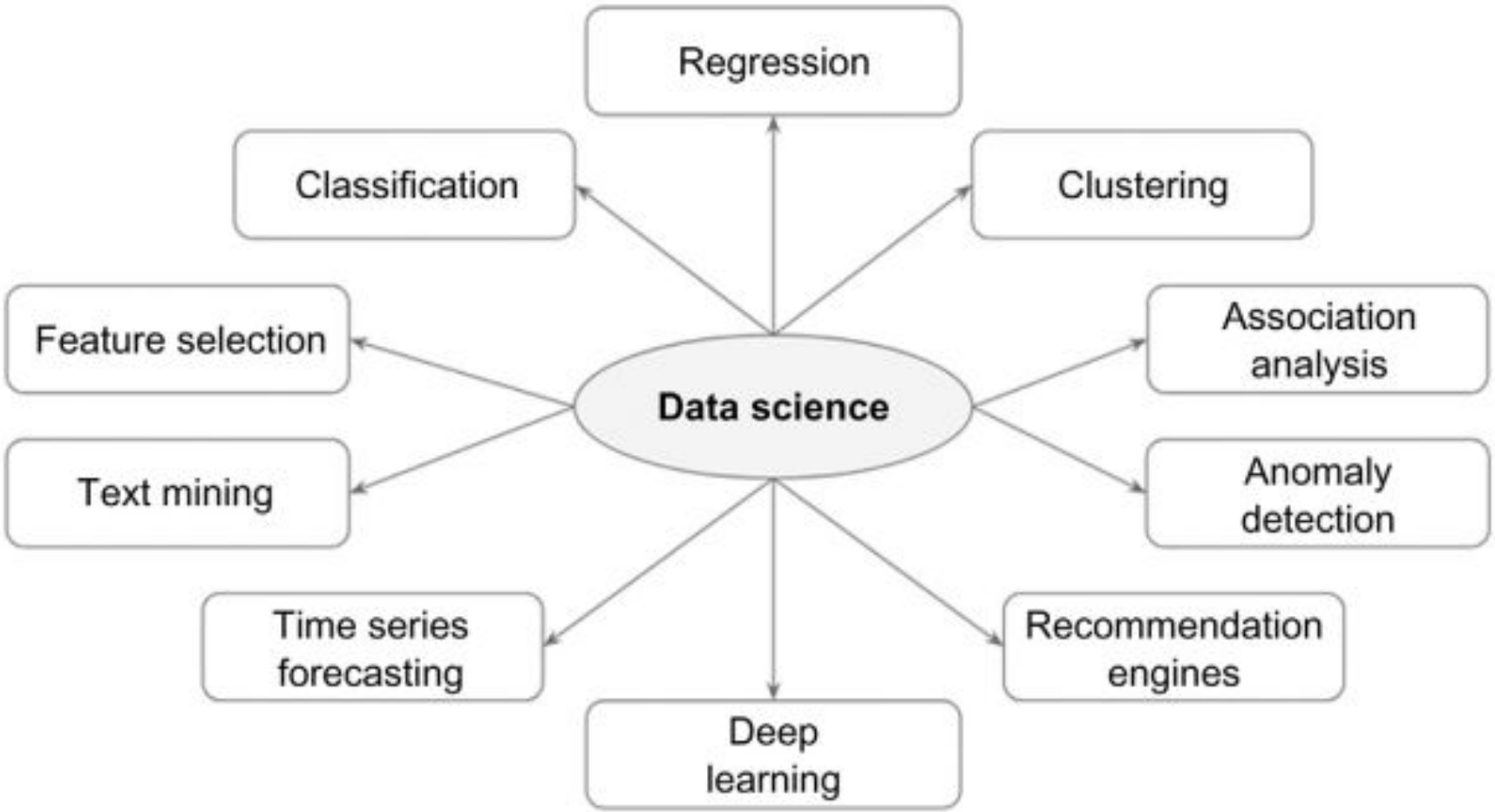



**FIGURE 1.1**
Artificial intelligence, machine learning, and data science.

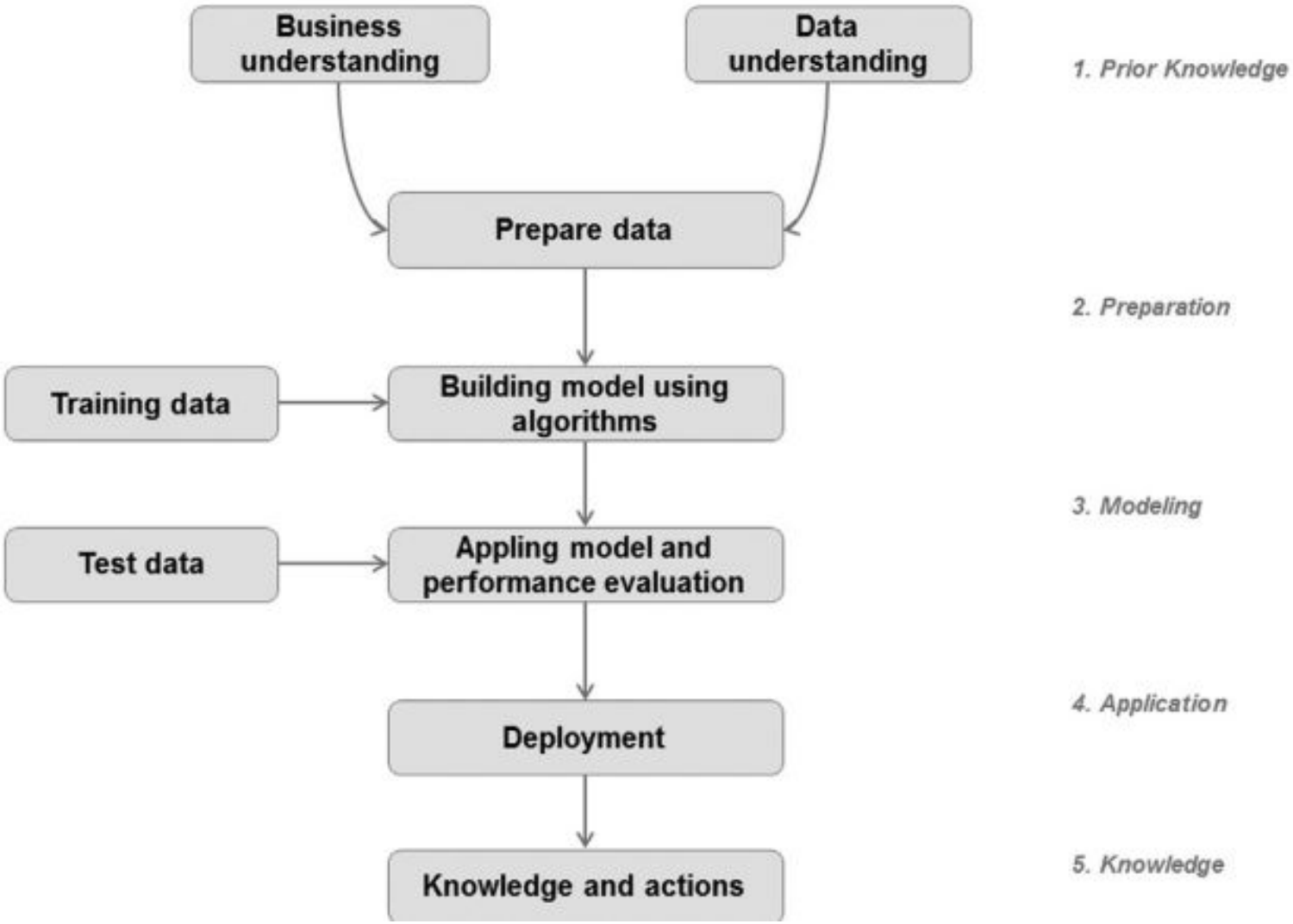
# Data Science Models



# Data Science Tasks



# Data Science Process



# Objectives of Data Science

- The data science process starts with a need for analysis, a question, or a business objective. This is possibly the most important step in the data science process (Shearer, 2000).
- Without a well-defined statement of the problem, it is impossible to come up with the right dataset and pick the right data science algorithm. As an iterative process, it is common to go back to previous data science process steps, revise the assumptions, approach, and tactics.